Sodium Chloride (NaCl) as Alternative Prevention and Treatment Against Trichodiniasis for Pike Perch (*Sander lucioperca*) in Recirculating Aquaculture Systems (RAS)

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Background

Pikeperch is a high valued species and a favorite of food fishes in Europe beside a salmon. Due to declining wild catches and increasing demand, there is a growing interest in intensifying production techniques using RAS. However, RAS comes with its drawbacks, such as the increased risk of disease transmission and ectoparasites due to high stocking densities[1]. These agents can be critical threats to the commercial success of aquaculture[3]. There are many broad-spectrum prophylactic agents for parasite treatment, but the use of low concentration of NaCl salt can be low cost and is one of the most harmless treatment under closed aquaculture systems[2][4].

Objective

To examine the effect of different NaCl salt application (0, 3, 6 PSU) on the prevalence and density of Trichodinids of naturally infected pike perch transferred into RAS

Methods

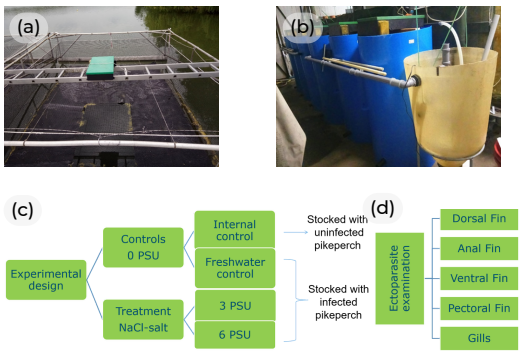


Figure 1 :
a. Overwintering in net cages for nine months: exposing pike perch to winter conditions that allow ectoparasite infection.
b. Pike perch (MBW 120.5±15.7g) are transferred into RAS. Each RAS: 57 fish, 3 rearing tanks (320 L) and a filtration unit (120 L)
c. Experimental design with four treatments
d. Ectoparasite examination in 5 sample locations on pike perch

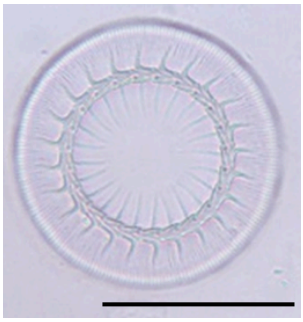


Figure 2. Image detail of a representative of the family of Trichodinidae under a light microscope originally taken at 40x magnification. Scale bar equals 100 µm

Results

After 21 days of observation, there was no difference in fish culture performance that includes parameters as initial and final bodyweight, SGR, and FCR. There was no mortality observed during the trial.



Trichodinid prevalence

At the beginning of the trial, all pike perch were infected with trichodinid. In contrast, there was no Trichodinid found in the internal control. The prevalence or infection rate of trichodinid in the freshwater control remained unchanged as the experiment progressed. Interestingly, a sharp reduction (by 83.3%) was observed in the 3 g NaCl/L treatment, and no trichodinids were detected in the 6 g NaCl/L (Figure 3).

Trichodinid density

There was no significant difference of the average trichodinid density per viewing field and fish (TDVF) between freshwater control, 3 g NaCl/L and 6 g NaCl/L at the beginning of the culture. However, the TDVF in the 3 and 6 g NaCl/L groups were decreased significantly (71.4% and 83.8% reduction) than the freshwater control (only 6.4%) after 21 days of treatment (Figure 4).

Trichodinid composition (%) on different body sites

The highest share of trichodinid densities was found on the gill arch of the fish (35-45%) followed by the back (26-40%), the ventral (13-23%), anal (5-8%) and pectoral fin (3-4%) (Figure 5).

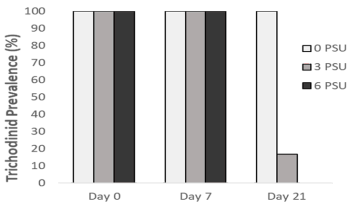


Figure 3. Average Trichodinidae density per viewing field and fish (TDVF) on individual pike perch reared in freshwater RAS with different NaCl treatments at day 0, 7 and 21 of the experiment. 12 fish were examined for ectoparasites per sampling. Significant differences between the treatments at any time point are indicated with a letter.

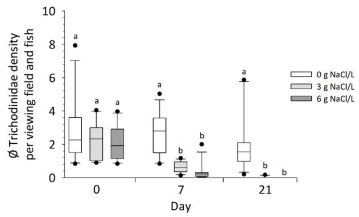


Figure 4. Average Trichodinidae density per viewing field and fish (TDVF) on individual pike perch reared in freshwater RAS with different NaCl treatments at day 0, 7 and 21 of the experiment. 12 fish were examined for ectoparasites per sampling. Significant differences between the treatments at any time point are indicated with a letter.

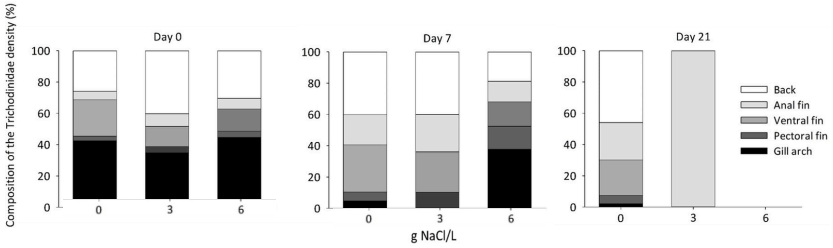


Figure 5. Composition of the trichodinid density (%) based on the five sample locations on pike perch reared in freshwater RAS with different NaCl treatments at day 0, 7 and 21 of the experiment. 12 fish were examined for trichodinids per sampling.

Conclusions

- The continuous application of 6 g/L NaCl can reduce Trichodinids to zero on pike perch with a body weight of 120 g under RAS conditions without additional medications and disinfectants.
- In summary, NaCl can be considered an effective and low-cost alternative with the potential to replace conventional therapeutics in pike perch aquaculture.

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